PAPER SUMMARY

AION Agent Harness: An Inference-Aware Self-Improving Agent Harness for Confidence-Gated Multi-Step Task Execution

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ABSTRACT

Current agent harnesses treat the underlying inference engine as an opaque token source, receiving outputs without any signal about the model's internal confidence. This opacity prevents harness-level intervention when a reasoning step is uncertain, allowing errors to propagate silently through multi-step pipelines.

We present AION Agent Harness, an inference-aware agent harness that crosses the harness/inference boundary by consuming the confidence score C(s) produced by the AION inference framework at each reasoning step. AION Agent Harness introduces three novel mechanisms: (1) Confidence-Gated Step Execution, which routes low-confidence steps to specialist adapters before continuing the pipeline; (2) Error Propagation Containment, which annotates downstream steps with upstream uncertainty signals to prevent silent compounding errors; and (3) Harness State Updates, where the Harness Feedback Controller rewrites step prompts between agent runs based on per-step confidence outcomes.

HIGH-LEVEL ARCHITECTURE

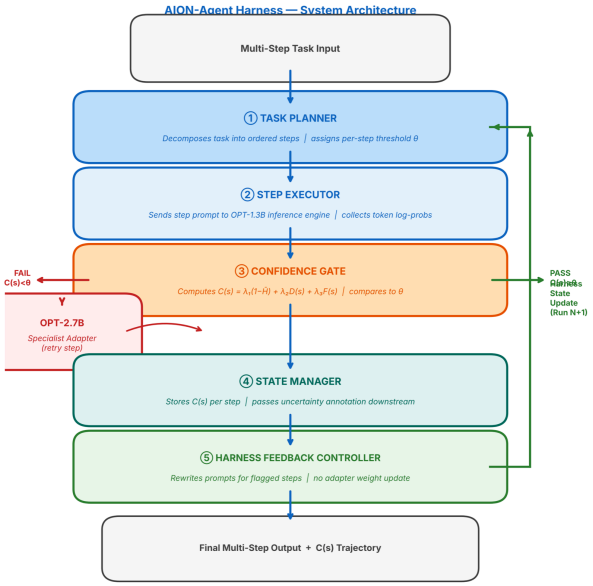


Fig. 1. AION Agent Harness system architecture: Task Planner, Step Executor, Confidence Gate, State Manager, and Harness Feedback Controller.

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